



Contents lists available at ScienceDirect

Journal of King Saud University – Computer and Information Sciences

journal homepage: www.sciencedirect.com

Software quality: Application of a process model for quality-in-use assessment

Leonice Souza-Pereira^{a,b,*}, Nuno Pombo^{a,b}, Sofia Ouhbi^c^a Universidade da Beira Interior, Covilhã, Portugal^b Instituto de Telecomunicações, Lisboa, Portugal^c Department of Computer Science and Software Engineering, CIT, UAE University, UAE

ARTICLE INFO

Article history:

Received 8 February 2022

Revised 9 March 2022

Accepted 31 March 2022

Available online 21 April 2022

Keywords:

Evaluation Model

Quality in use

Quality measures

Software quality assessment

ISO-IEC 25022

Clinical Decision Support Systems (CDSS)

ABSTRACT

Quality-in-use characteristics can be measured to assess the quality of the software and even point out problems in its production since they reflect the quality of the system perceived by the end-users identified in the running systems. Measuring such characteristics is a way of verifying that the software meets the requirements and expectations of the interested parties.

This study presents the validation of a process model for the evaluation of the quality-in-use characteristics of a system. The model works with the identification and measurement of important characteristics of the system to determine the quality of the software, and thus identify potential problems that may inhibit its use.

We applied our process model in a clinical decision support system in the oncology field using a questionnaire for gathering data for the measurement. The questionnaire mainly sought to identify user satisfaction with the use of the software, and the answers permitted us to measure the quality-in-use characteristics of the system.

Our experiments revealed the adequacy of our evaluation process model and the evaluated CDSS reached a score of 71.35, which is advantageous according to our standard of measures and meets the expectations of users.

© 2022 The Author(s). Published by Elsevier B.V. on behalf of King Saud University. This is an open access article under the CC BY-NC-ND license (<http://creativecommons.org/licenses/by-nc-nd/4.0/>).

1. Introduction

External quality attributes of a software, such as maintainability and usability, are affected by subjective factors, such as user experience (UX) (Sommerville, 2016), knowledge, and digital literacy. In addition, usability can be defined by the extent or the level “to which a system, product or service can be used by specified users to achieve specific objectives with effectiveness, efficiency and satisfaction in a specified context of use” (ISO/IEC, 2019; ISO/IEC, 2011). Indeed, the degree in which a software system satisfies specific requirements, as well as users’ needs and expectations, refers to the quality of the software product (Miguel et al., 2014).

With these principles in mind, to have an effective evaluation and guarantee of the software quality, it is necessary to follow models that may describe and track the quality, during the software development cycle, and in the course of its subsequent maintenance activity (Arnican et al., 2020).

Thus, international software quality standards emerged to guide the achievement and measurement of this quality throughout concepts, procedures, and metrics. The recent series ISO/IEC 25000 - System and Requirements for Software Quality and Evaluation (SQuaRE) standard (ISO/IEC, 2019) includes models and frameworks that could be used to assist software developers in evaluating the quality of their products.

One of these models is the Quality in Use (QiU) model, composed of multiple characteristics (ISO/IEC, 2011), and other one which offers methods and metrics to measure that QiU (ISO/IEC, 2016).

According to Djordjevic (Djordjevic, 2017), product quality attributes are the cause, and QiU attributes are the effect. Whence, measuring and evaluating the QiU characteristics is a way to confirm the external quality of the software (measurement of behaviour), which can verify its internal quality (static measurement of intermediate products). Measuring the QiU characteristics allows identifying system functionalities that must be improved to increase its quality (Pincioli, 2016).

Since the usability is considered the cornerstone of the UX (Evans et al., 2020; Hassan and Galal-Edeen, 2017; Atoum and Bong, 2015) then is quite similar to measure any of them. In fact,

* Corresponding author.

E-mail addresses: leonice.pereira@ubi.pt (L. Souza-Pereira), ngpombo@di.ubi.pt (N. Pombo), sofia.ouhbi@uaeu.ac.ae (S. Ouhbi).

the UX partially reflects the QiU of the system because it is a major parameter of user acceptance and satisfaction when using software. In addition, the QiU assumes extreme relevance when the software is operated either by non-digital users or users whose background or experience differs from the computer science umbrella. Congruently, scenarios in which the decision-making relies on software, such as healthcare or industry, are highly dependant on the QiU of the system.

Into the healthcare context, practitioners use computer systems to support them in the clinical decision-making, in terms of awareness and better understanding of their conditions (Souza-Pereira et al., 2020).

However, these systems, a.k.a. Clinical Decision Support Systems (CDSSs) challenge for its adoption due to a myriad of causes, such as: (1) end-user's perception of the system, (2) system usefulness, and (3) reliability of system outputs (Alshare et al., 2019; Khalifa, 2014; Zikos, 2017).

Therefore, measuring the quality characteristics of a CDSS can allow us to identify reasons for non-acceptance of CDSSs, since the values found can point to deficiencies and weaknesses in the system, which affect its usability. The measurement also permits to identify the non-compliance with user requirements, which harms UX, and it can even point to problems in the system's development.

In this paper, we have the ambition to validate a process model for quality in use assessment of a CDSS, developed in our previous work (Souza-Pereira et al., 2021b), into a CDSS context and present results and lessons learned.

Our proposed process model, based on the Goal-Question-Metric (GQM) method and following the international standards ISO/IEC 25010 and ISO/IEC 25022, aims to identify deficiencies in a CDSS by measuring QiU characteristics and, thus, contribute to the increase of the acceptance of the software by the medical community. This paper is organised as follows. In Section 2 we present background information and definitions required to understand our proposed evaluation process model and its validation.

The preparation for validating the proposed model is presented in Section 3, through the steps for choosing the CDSS to be evaluated and applying the model. Section 4 presents the application of the proposed process model. Section 5 discusses results of the measurement. The conclusion, limitations of the study, and future work are presented in Section 6.

2. Background

2.1. Software quality models

ISO/IEC 25000, or SQuaRE, (ISO/IEC, 2019), is a series of standards formed to create a framework for evaluating the quality of software products. SQuaRE is split into five divisions. We addressed two of them in our previous studies (Souza-Pereira et al., 2021a, 2021b): (1) Quality Model Division - ISO/IEC 2501n, and (2) Quality Measurement Division - ISO/IEC 2502n.

ISO/IEC 25010 - System and software quality model (ISO/IEC, 2011) is part of the ISO/IEC 2501n, together with the data quality model - ISO/IEC 25012. It comprises two main subdivisions:

1. **The software product quality model**, which includes eight characteristics and relates to the static properties (i.e., internal quality attributes), and dynamic properties (i.e., external quality attributes).
2. **The Quality in Use model**, which includes five characteristics and considers the user's interaction with the software in a context.

ISO/IEC 25022 - Measurement of Quality in Use (ISO/IEC, 2016) is the SQuaRE's standard that normalises the measurement

of QiU characteristics of a software product. It comprises a series of measurements associated with the QiU characteristic (Efficiency, Effectiveness, Satisfaction, Freedom from risk and Context Coverage), including methods and mathematical functions that are applied for measurement, in addition to instructions for normalisation and interpretation of these measurements. It also presents a guideline for evaluating the QiU of a software system.

2.2. GQM method

The GQM method, which was developed by Basili and Weiss (Basili and Weiss, 1984), includes four phases: (1) The planning phase, where the project to be measured, is selected and defined; (2) the definition phase, where the measurement goal, corresponding questions, employed metrics, and hypotheses are documented; (3) the data collection phase, and (4) the interpretation phase, where the collected data are transformed to measurement results by applying predefined metrics.

Several studies have employed the GQM approach (Bukhari et al., 2018; Sacha, 2006; Tsuda et al., 2019; Alves et al., 2016; van Solingen and Berghout, 1998), adapting it to match their proposals, simplifying it, or using it to develop their own methods. We adapted the GQM to select measures to assess QiU characteristics.

3. Methodology

This section presents the steps considered for the application of our proposed QiU evaluation process model, from the choice of the CDSS to the measurement of the characteristics.

3.1. The CDSS's choice

The initial idea was to apply our proposed process model in a hospital, working directly with health professionals, with a CDSS already in use by them. Unfortunately, this became impossible due the Covid-19 pandemic situation, which affected and changed all situations that could be considered normal, especially within a medical-clinical environment. Thus, the solution was to apply the assessment in a free CDSS, in a non-restrictive way. We used the Google Store to select a CDSS that best met our expectations in order to identify potential usage scenarios and select QiU characteristics for the measurement, which in our case study was limited to Satisfaction and Efficiency QiUs.

3.2. The gathering process

The gathering process took place from the 2nd to the 5th of June 2021, with no new apps submitted after this date being considered. We found 248 apps that were carefully studied and classified into groups in order to identify the software's purposes.

CDSS was classified within its main proposal as:

1. CDSS;
2. educational software (service protocols, videos, glossaries, procedure flowchart, technical-scientific content, use case), guidelines, interpreters, chats, e-books software, guidelines, interpreters of clinical definitions;
3. clinical calculators (the ones with only this type of functionality),
4. Medical Health Record (MHR), Personal Health Record (PHR), workout apps, self-care app;
5. agendas or outside the clinical context and others.

Fig. 1 presents the selection process of the CDSS used in this study.

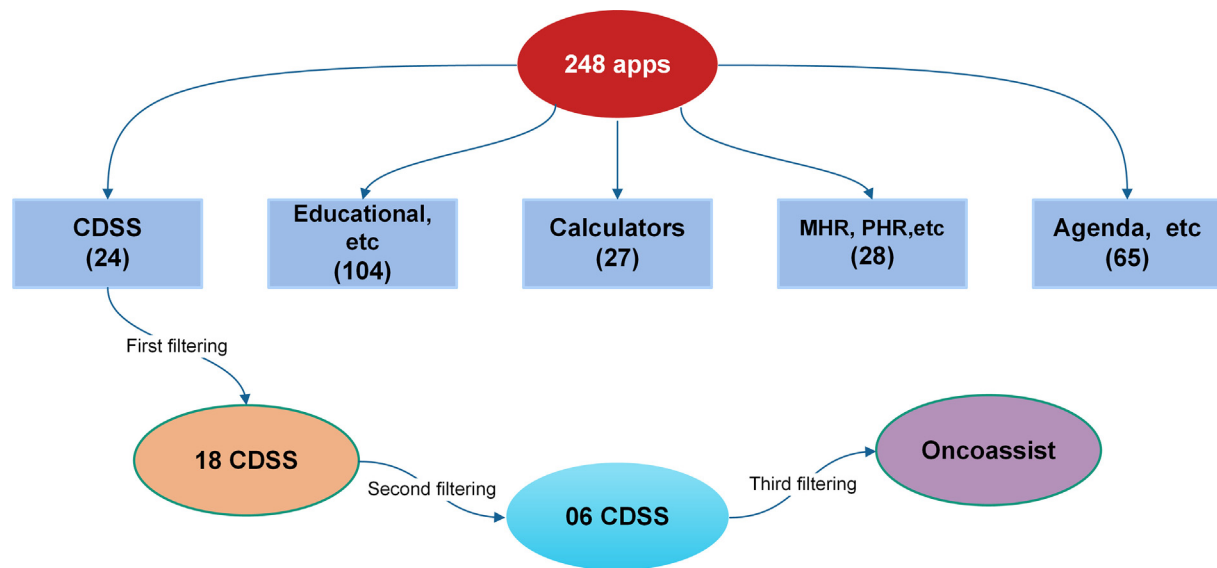


Fig. 1. Found Apps and the filtering process.

After this classification, 24 CDSS were identified, which were submitted to three filtering criteria.

First filtering: We considered the CDSS that underwent updates between the year 2018 and June 2021. This reduced our selection of CDSS to 18 systems.

Second filtering: The downloads of these CDSS should comply with a quantity criterion based on the year of publication:

1. year of publication 2018, should have a number of downloads equal to or greater than 5,000 (0 CDSS).
2. year of publication 2019, should have a number of downloads equal to or greater than 1000 (0 CDSS).
3. year of publication 2020, should have a number of downloads equal to or greater than 500 (1 CDSS).
4. year of publication 2021, should have a number of downloads equal to or greater than 100 (5 CDSSs).

After the second filtering, 06 CDSSs were selected from the 18 coming from the first filtering.

Third filtering: We present in Table 1 questions to be answered by the 6 CDSSs identified in the previous step, after being executed and tested for analysis and collection of characteristics. We classified them according to the number of positive answers. This step was not concerned with the specific functionalities of CDSS.

The CDSS with the most positive responses was **Oncoassist**¹, a CDSS of the oncological area, answering 8 “yes” to the 9 questions, which was chosen. The classification of CDSS is presented in Table 2. After choosing CDSS to be evaluated, we move on to applying our proposed measurement process model.

3.3. Process model for evaluation of QiU

This study aims to present the validation of our process model for the evaluation of QiU characteristics for CDSSs. This model, exhibited in Fig. 2, consists of five main phases:

1. Identification of scenarios and context of use: this phase identifies the contexts of use of a system, in our case study, of a CDSS. Here, the scenarios where the main system functions are performed are pointed out;

Table 1
Filtering Questions.

Id.	Question	Positive answers
1	Does the app provide information about updates in the app?	3
2	Can we suggest changes, improvements or additional features in the app itself?	2
3	Can we send email to support/developer team?	5
4	Does the app offer an official support by the Doctors?	6
5	Does it present terms of use, or privacy Policy, or disclaimer?	5
6	Does the app target at a specific disease or area?	4
7	Does it present the possibility to save or send results?	1
8	Does it present a final report with the suggestion/ decision?	5
9	Does it present a theoretical basis to support results?	6

Questions applied to the 06 CDSS and the number of positive answers.

Table 2
CDSSs classification after third filtering

Id.	CDSS	Number of positive answers
1	Oncoassist	8
2	Neuro Mind	7
3	TherapySelector	7
4	CFS	5
5	MDCalc	5
6	Calculate	5

CDSSs classification by number of positive responses.

2. Selection of measures, methods and metrics: in this phase, the selection of the characteristics and sub-characteristics of QiU to be evaluated, as well as the choice of the quality measure to be measured, takes place. Such measures are classified as **general**, when they are generally applicable and used in a wide range of situations, and **specialised**, when they are used for specific needs;
3. Quality in use measurement: in this phase, the mathematical function associated with the chosen quality measure is applied through tests with users and data collection of the scenario(s) selected by the evaluator;

¹ oncoassist.com

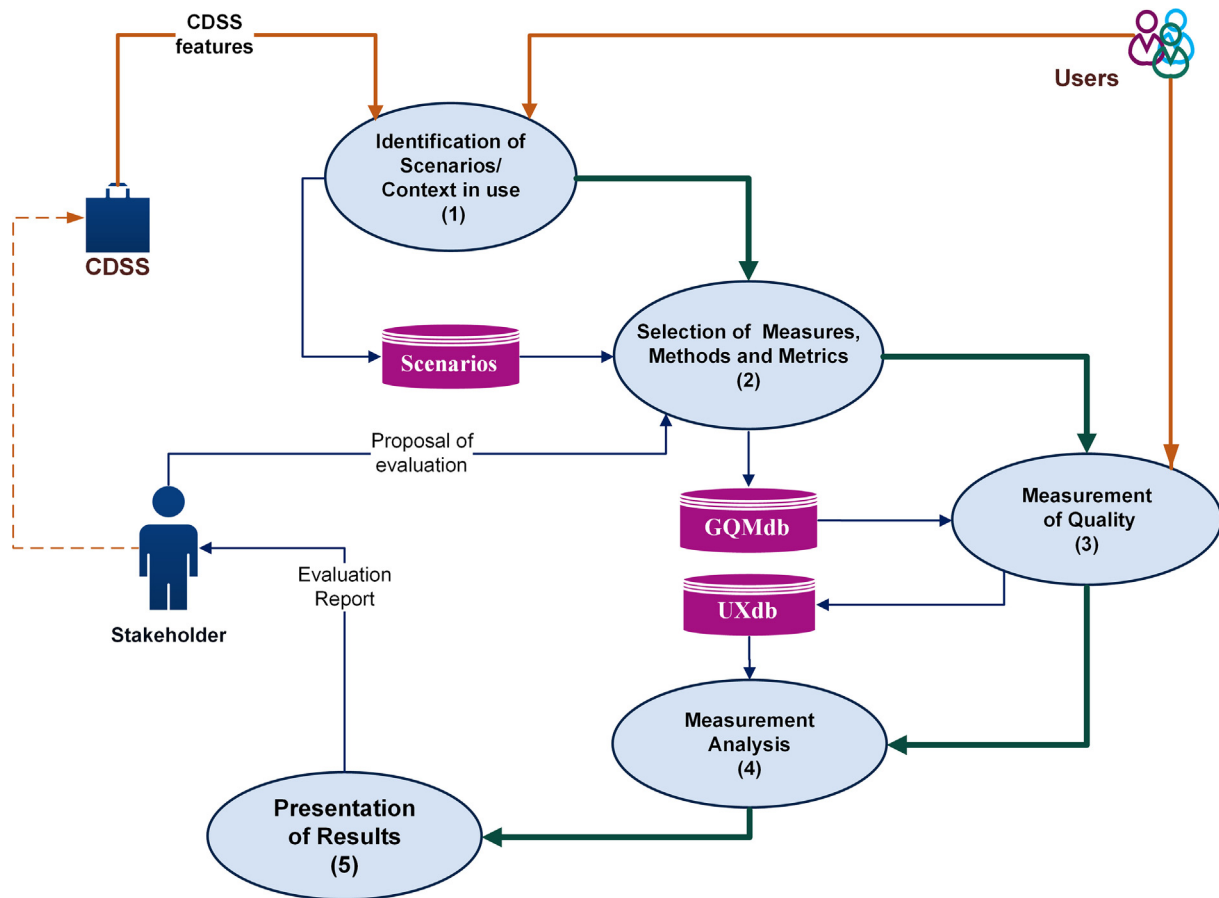


Fig. 2. Process model for QiU evaluation.

4. Measurement analysis: the results are analysed after measuring the characteristics. In this step, we have the answers to the questions (Q) of the GQM method as well as the interpretation of the measurement results of the mathematical functions;
5. Presentation of results: In this phase, the results obtained by the interpretation of the measured values are presented. It is suggested that the presentation is through a measurement report, where the strengths and weaknesses of the CDSS and graphic images are presented for better and easier visualisation.

These phases are described in our study (Souza-Pereira et al., 2021b) in detail, presenting the QiU characteristics and measures to assess them as well.

4. Proposed model application

The following subsections bring this application and comprise the four first phases of our model. The fifth phase is described in Section 5.

4.1. Identification of Scenarios and Contexts-in-Use on the CDSS

We have identified 21 functionalities in the CDSS Oncoassist and 18 outcomes. Table 3 presents the functionality-outcomes relation. The number of candidate scenarios is related to the functionalities of the CDSS and outcomes produced by them, as well as the number of users running the system during quality assessment (Souza-Pereira et al., 2021b).

We had no idea how many participants we would have in the validation process of the proposed model. So, there was no way

to know in advance how many scenarios we would have since it was related to the number of users.

Furthermore, given the circumstances of measuring quality, it would be impractical to address all QiUs and all scenarios.

So we limited ourselves to the Satisfaction and Efficiency QiUs for our initial idea of applying the model. We also limited in addressing the general use of CDSS and in two specific functionalities, identified in Table 3, to compose our scenarios. These functionalities were, namely, the **creation of access credentials** (FO1) and the use of the functionality - **Adjuvant Tools >>Breast Cancer** (FO14).

4.2. Selection of Measures, Metrics and Methods

Once we identified the scenarios and predefined the Satisfaction and Efficiency QiUs characteristics, the next step was to select the measures, metrics and methods to achieve the measurement of these characteristics.

Our proposed model adapted the GQM method to create a list of requests/responses related to CDSS functionalities to be evaluated, and to CDSS in general. This list is presented in Table 4. Based on this GQM list, we drew up an online questionnaire as a method to get information for the assessment of QiU of the CDSS.

Surveys or questionnaires are suggested by ISO/IEC 25022 (ISO/IEC, 2016) as a method to achieved the measurement and also applied in several studies (Kitchenham and Pfleeger, 2008; Lund, 2001; Lewis, 1995; Ives et al., 1983; Pavlič et al., 2018; Bauk et al., 2014; Nakai et al., 2016; Jung et al., 2004; Lew et al., 2012; Chen and Lee, 2009). For the creation of our questionnaire, we based on these studies and also in examples of enterprises focused

Table 3
Functionalities-Outcomes

ID	Functionality	Outcome
F01	Account » Sign up	User account created
F02	Account » Settings » Logout	Quit of the system
F03	Account » Settings » Change country	User labor country details
F04	Account » Settings » Edit Profile	User data changed
F05	About » Settings » About » Oncoassist	Information about the authors and about the app are shown
F06	About » Settings » Terms and conditions	End User License Agreement are shown
F07	About » Settings » Privacy Policy	Personal data collected and the privacy policy are shown
F08	About » Settings » Feedback	App improvement or report incidents are suggested
F09	About » Settings » Tell a friend	App link are sent
F010	Favourite	Favourite a page/ functionality
F011	Search	Search execution
F012	Adjuvant Tools » Colon Cancer	Estimated survival rates
F013	Adjuvant Tools » GIST	Estimated survival rates
F014	Adjuvant Tools » Breast cancer	Estimated survival rates
F015	Adjuvant Tools » Lung Cancer	Estimated survival rates
F016	Formulas	Formulas results
F017	Prognostic Scores	Calculated prognostic scores
F018	Toxicity Grading (CTCAE)	Toxicity results
F019	AJCC/TNM staging	AJCC
F020	Drug Interaction Checker	Anti-cancer drug interaction
F021	Advance Breast Cancer tools	Line of treatment in different clinical trials (BOLERO, PALOMA, Manarch, SOLAR, MONALEESA)

Functionalities and their outcomes of the Oncoassist CDSS.

on user experiences and research expertise (Nielsen, 1994; Nielsen, 2020).

A guideline explaining the target scenarios to be evaluated was distributed to the participants, instructing them to use the system in its entirety in order to get feedback on the system's overall usability. So we could subtract your satisfaction from your answers when using the app.

According to ISO/IEC 25022, international standard used in our study, each measure is related to a function and a method to apply to obtain this measure (ISO/IEC, 2016). Table 5 presents the measures and mathematical functions used in our measurement.

4.3. QiU Measurement and Analysis

The questionnaire was applied from October 16Th to November 2Nd. We distributed it to health professionals, students of medicine, bio-medicine, biomedical engineering, and computer engineering of software quality area. Only users who had not registered on the Oncoassist platform had participated in the quality measurement process, because one of the questions concerned the access credentials creation. We obtained 18 completed questionnaires, whose answers were grouped and measured using selected mathematical functions from the ISO/IEC 25022 standard, according to each characteristic and sub-characteristic evaluated.

Several studies were used as a support for the analysis and summary of the results, where it was possible to observe the use of weights for the characteristics (IBM-Corporation, 2014; Sulla-Torres et al., 2020), scores (IBM-Corporation, 2014; Chen and Lee, 2009), System Usability Scale (SUS) (Bangor et al., 2008; MeasuringU, 2021) and also statistical methods (Jung et al., 2004). We based on those studies to elaborate our range of values to comparisons, considering acceptable levels of quality (gold-standards) to decide if the CDSS is suitable for the intended purpose.

We convert the range of values to a scale of 0–100 and consider the results.

(1) excellent if above 90.0, (2) between 80.0 and 90.0 is very good, (3) between 65.0 and 80.0 is good, (4) between 60 and 65 is reasonable, (5) between 51 and 60 we consider bad and (6) below 51, awful.

We summarised the measures using the average (simple or weighted) or simple sum of the results of mathematical functions

of the ISO/IEC 25022 standard, and compare the result of each measure with a score considered standard.

In the end, we were able to specify the value for each question and get the overall value for each QiU characteristic and sub-characteristic, and subsequently classified it as satisfactory or not in the context of the CDSS.

5. Presentation of results

The measurement and analysis of the found values were presented separately: (1) considering the specific functionality, and (2) the CDSS in general. In both were considered (a) each sub-characteristic measure, (b) each characteristic measure, and (c) overall measure.

Within this division, the Efficiency and Satisfaction QiUs were also separately considered. For Satisfaction characteristic assessment, 3 sub-characteristics were measured in the specific functionality part, and 4 sub-characteristics for the general part, as can be observed in Table 4. Finally, we presented the final classification of the CDSS. Two functionalities of the Oncoassist CDSS were chosen to be evaluated: Adjuvant tools » Breast Cancer and Creation of access credentials.

5.1. Adjuvant tools functionality results

The Adjuvant Tools feature » Breast Cancer was rated by participants as a good feature, with 77% positive responses about its usefulness in aiding decision making, a score of 78% about visual aids being important for a good understanding of the presented results, and 83% about the feature being easy to use. These are measures from the Trust (the first two) and (UX) Pleasure (the last) sub-characteristics. The majority of participants took less than 6 min to execute the functionality and to understand the results presented. Fig. 3 shows the time spent in this task.

In the context of this functionality, we present in Fig. 4 measurement of the efficiency and satisfaction characteristics through their sub-characteristics measures.

Two participants (11%) complained about the functionality related to the layout (badly arranged buttons, field size, not fixed menu bar), and not using the keyboard to fill in the data. Therefore, we can deduce the evaluated functionality was accepted and well-rated by the users, with an average measurement of 70,42%.

Table 4
The GQM list.

Section	Question	Charact.Subcharac
Personal	How do you classify your knowledge on Oncology	N/A
	What platform did you use to access the software?	N/A
	How long did it take you to run the functionality, from data entry to understanding the result?	Efficiency
	In your opinion, how do you classify the usefulness of this functionality to help decision making?	Satisfaction-Trust
Adjuvant Tools Breast Cancer	Visual information, graphics and charts presented on the results are important for my understanding.	Satisfaction-Trust
	The presentation of the icons is important for understanding the results.	Satisfaction - (UX)-Pleasure
	The functionality is easy to use.	Satisfaction - Usefulness
	Data entry is made easy.	Satisfaction - Usefulness
	The help option, next to each field to be filled out, is useful for using the functionality.	Satisfaction - Usefulness
	The results presentations are explanatory	Satisfaction - Usefulness
	You believe you would use this functionality frequently to help me in decision making.	Satisfaction - Usefulness
	Do you have any complaint or suggestion regarding the functionality Adjuvant Tools >>Breast Cancer?	Satisfaction - Usefulness
	How long did it take you to create your account?	Efficiency
	How do you classify your experience in the account creation	Satisfaction - (UX) - Pleasure
	How do you rate your experience using the software?	Satisfaction - (UX) - Pleasure
	The software is a useful application in making support medical decisions (rate)	Satisfaction - Trust
	The software seems trustworthy to you	Satisfaction - Trust
	The layout of software is well done.	Satisfaction - Usefulness
	The results presentation are clear and objective.	Satisfaction - Usefulness
	References and the additional explanations to support the results are important and useful.	Satisfaction - Usefulness
	The software has important functionalities.	Satisfaction - Usefulness
	This software is easy to remember.	Satisfaction - (UX) - Pleasure
General	You feel comfortable using the software.	Satisfaction - (UX) - Pleasure
	This software is easy to use.	Satisfaction - (UX) - Pleasure
	Your overall satisfaction with using this software is high.	General
	You believe you would use this software frequently to help you in decision making.	Satisfaction - Usefulness
	Do you have any complaint or suggestion to be done about the software?	Satisfaction - Usefulness

The questions of GQM list and the QiU to be measured.

Table 5
Measures and mathematical functions.

Charac.	Subcharac.	Measure	ID	Function	Description
Efficiency	N/A	Task Time	Ey 1-G	$X = T$	T = Time taken to successfully complete a task
Satisfaction	Trust	User Trust	STr 1-G	$X = A$	A = Psychometric scale value from a trust questionnaire
	(UX) Pleasure	User pleasure	SPl 1-G	$X = A$	A = Psychometric scale value from a pleasure questionnaire
	Usefulness	Satisfaction with features	SUs 2-G	$X = \sum(A_i)$	A_i = Response to a question related to a specific feature
			SUs 4-G	$X = \frac{A}{B}$	A = number of users using a particular feature; B = number of users in an identified set of users of the system
			SUs 6-G	$X = \frac{A}{B}$	A = number of user complaints for a particular feature; B = Total number of user complaints about features
			SUs 3-G	$X = \frac{A}{B}$	A = number of users using a particular feature; B = number of potential users who could use the particular feature
			SUs 5-G	$X = \frac{A}{B}$	A = number of users complaining; B = number of users using the system
	General	Overall Satisfaction	SUs 1-G	$X = \sum(A_i)$	A_i = Response to a question

Functions used to measure the QiU characteristics of the CDSS (according to ISO/IEC, 2016).

5.2. CDSS general measurement results

Although it was considered a specific functionality to be measured, the results' presentation of the creation of access credentials was integrated into the group of general measures of the CDSS, to be according to our GQM list and our distribution of questions in the applied questionnaire.

Therefore, about the Creation of access credentials functionality, some users took over 3 min in this task, which reduced the efficiency rate in the task execution. The time spent to create the account to access the system is presented in Fig. 5 with a score of 51,2%.

Considering the whole system, the efficiency and satisfaction characteristics measures are shown in Fig. 6, with all satisfaction sub-characteristics scored above 65%.

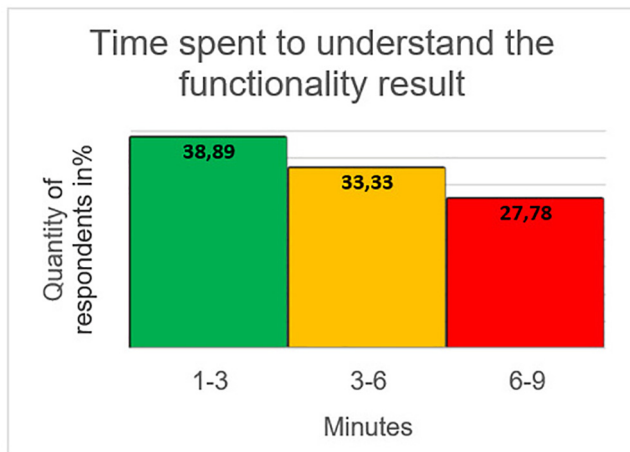


Fig. 3. Task time measurement.

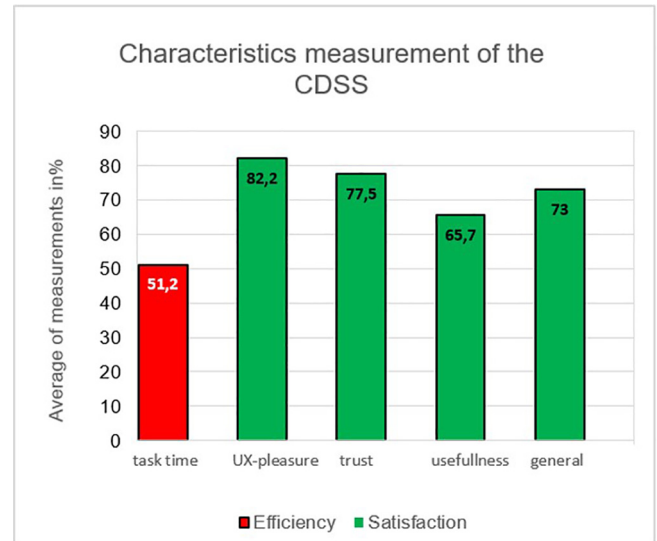


Fig. 6. Measures of efficiency and satisfaction characteristics of the CDSS.

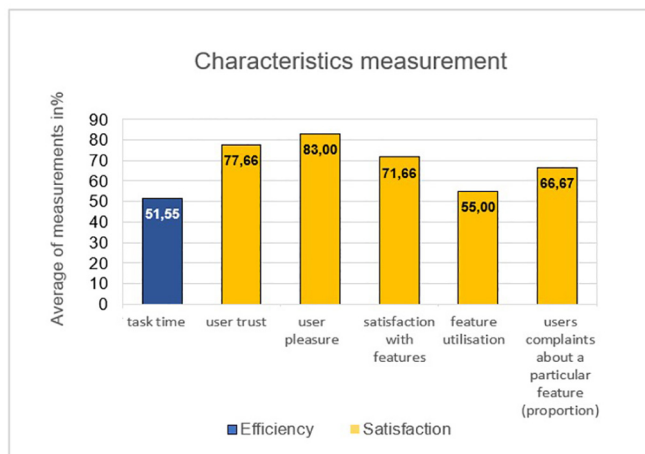


Fig. 4. Measures of efficiency and satisfaction characteristics of the functionality Adjuvant tools »Breast Cancer.

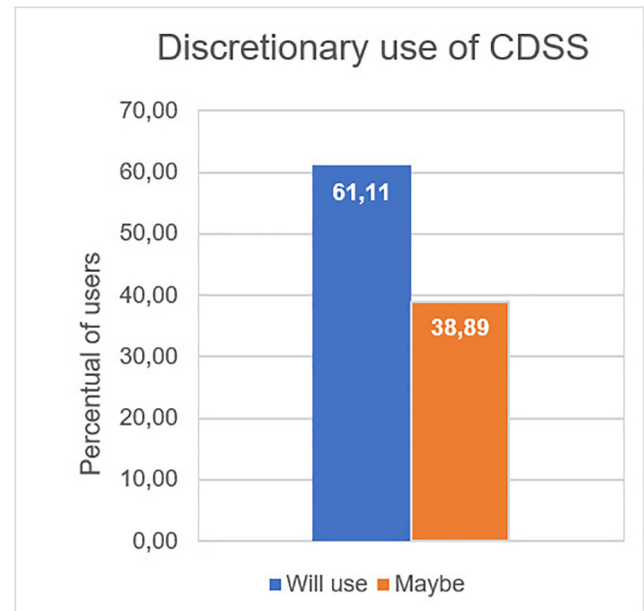


Fig. 7. CDSS usage.

average value of the CDSS general measurement, considering both functionalities Satisfaction and Efficiency, was 72,28%.

5.3. Final results

Measuring the QiU characteristics of the CDSS, considering the scenarios where two specific characteristics were assessed and also the system as a whole, we found a significant value of 71.35%, being the CDSS classified as GOOD according to our scale of values, and meeting the needs and expectations of users. The found values are presented in Fig. 8.

6. Conclusion, limitations and future work

In this paper, we validated a process model for the evaluation of quality in use of a CDSS, following the ISO/IEC 25022 and ISO/IEC 25010 standards. The proposed process model employed an

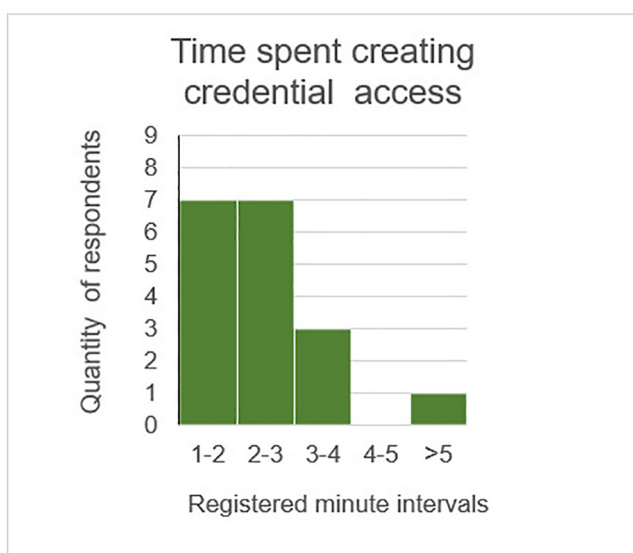


Fig. 5. Task time on credentials creation.

None of the participants ruled out the use of CDSS, with the majority (61,11%) claiming they could use it. This result is presented in Fig. 7, through the discretionary usage measure. The

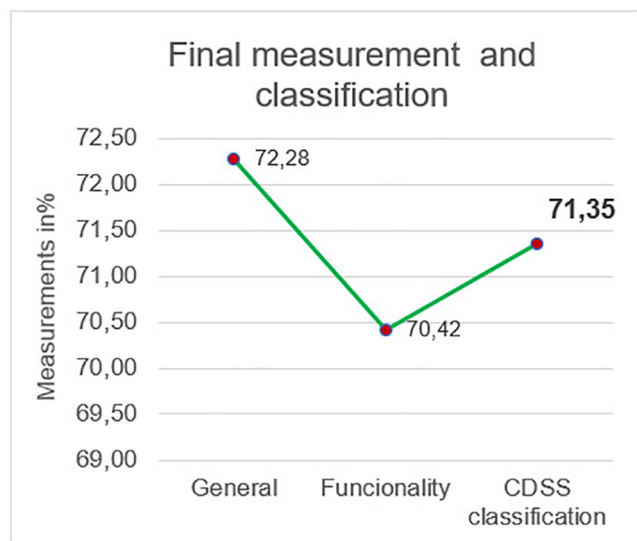


Fig. 8. Classification of the Oncoassist CDSS.

adapted GQM method to determine the measures and metrics to be used in evaluating the QiU characteristics, which in our case study were the Efficiency and Satisfaction characteristics.

We searched for a free-use CDSS to apply our process model and, firstly, we found 248 apps. These apps went through three judicious selection filters, and finally, we selected a CDSS: Oncoassist.

We applied this process model to Oncoassist CDSS through an online questionnaire created from our GQM list and distributed to healthcare professionals, software quality engineers, and students in the fields of medicine, biomedicine, and biomedical engineering.

The measurement of the QiU characteristics of the Oncoassist system reached 71,35% in our gold standard score range, clarifying that CDSS is a advantageous system and meets users' expectations.

We conclude that our proposed process model is useful to select and assess QiU characteristics, contributing to software quality professionals in the search for confirmation of the quality of a system in terms of meeting the needs and expectations of the user.

As a limitation of our work, we can mention the fact that the evaluation did not take place in a hospital environment, neither as a result of a daily CDSS usage by a medical team. These constraints, limited the adoption of additional efficiency measures than those employed in our online survey.

Besides this, we have not experienced the health professional's work routine with the CDSS, which could affect the measurement of QiU, based on the way the system had been used and perceived. Another detail that can be considered a deficiency is the number of responses obtained, which were 18 questionnaires answered. This can be considered a bias in the measurement and final classification of the system, even though studies have corroborated with the fact that, for measuring usability, few answers are enough to formalise reliable results. The standard ISO/IEC 25022 (ISO/IEC, 2016) brings that "...by controlling of the context of evaluation, experience has shown that reliable results can be obtained with a sample of only eight participants...", and Nielsen et al. (Nielsen and Landauer, 1993) demonstrated, using a mathematical model, that five is an ideal number of users to evaluate systems, as they found 75% of the usability problems. A lesson learned from this study was to realise that online surveys elucidate a low adherence by participants. Only eighteen people of the 76 invited professionals responded to our invitation. This corroborates the importance of the presence of the evaluator with the system user at the time of the evaluation.

As a challenge, we can mention the need to change this low engagement culture to this type of metric, perhaps through a more collaborative proposal, with the use of intelligent agents in the generation of questions from previous answers, in order to identify where one answer contradicts another, or even prevents it from happening.

For future work, we intend to apply the model to measure the QiU of a CDSS in use within a hospital environment, and use it also to measure all QiU characteristics, not just those that impact the end-user.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgement

Leonice Souza-Pereira would like to thank the IFTM - Instituto Federal do Triângulo Mineiro Campus Uberlândia Centro, Uberlândia, Brasil, for their support. This work is supported by FCT/MCTES through national funds and when applicable co-funded by FEDER—PT2020 partnership agreement under Grant UIDB/EEA/50008/2020. (Este trabalho é financiado pela FCT/MCTES através de fundos nacionais e quando aplicável co-financiado por fundos comunitários no âmbito do projeto UIDB/EEA/50008/2020); Co-financed by the Portugal 2020 Program (PT 2020), in the framework of the Regional Operational Program of the Center (CENTRO 2020) and the European Union through the Fundo Europeu de Desenvolvimento Regional (FEDER), under the Grant Centro-01-0247-FEDER-072632— "NomaVoy - Nomad Voyager".

References

- Alshare, K.A., Alomari, M.K., Lane, P.L., Freeze, R.D., 2019. Development and determinants of end-user intention: usage of expert systems. *J. Syst. Inf. Technol.* 21, 166–185. <https://doi.org/10.1108/JSIT-08-2018-0108>. URL: <https://www.emerald.com/insight/content/doi/10.1108/JSIT-08-2018-0108/full/html>.
- Alves, J.M., Savaris, A., Von Wangenheim, C.G., Von Wangenheim, A., 2016. Software quality evaluation of the laboratory information system used in the Santa Catarina state integrated telemedicine and telehealth system. *Proc. - IEEE Symp. Comput. Med. Syst.* 2016-August, 76–81. doi: 10.1109/CBMS.2016.51.
- Arnican, V., Borzovs, J., Nesaule-Erina, A., 2020. Do we really know how to measure software quality? *CEUR Workshop Proc.* 2620, 9–16.
- Atoum, I., Bong, C.H., 2015. Measuring Software Quality in Use: State-of-the-Art and Research Challenges. *ASQ Software Quality Professional* 17, 4–15. URL: <http://arxiv.org/abs/1503.06934>, arXiv:1503.06934.
- Bangor, A., Kortum, P.T., Miller, J.T., 2008. An empirical evaluation of the system usability scale. *Int. J. Hum. Comput. Interact.* 24, 574–594. <https://doi.org/10.1080/10447310802205776>.
- Basili, V.R., Weiss, D.M., 1984. A Methodology for Collecting Valid Software Engineering Data. *IEEE Trans. Softw. Eng.* SE-10, 728–738. <https://doi.org/10.1109/TSE.1984.5010301>.
- Bauk, S., Šćepanović, S., Kopp, M., 2014. Estimating Students' Satisfaction with Web Based Learning System in Blended Learning Environment. *Educ. Res. Int.* 2014, 1–11. <https://doi.org/10.1155/2014/731720>.
- Bukhari, Z., Yahaya, J., Deraman, A., 2018. A conceptual framework for metrics selection: SMEs. *Int. J. Adv. Sci. Eng. Inf. Technol.* 8, 2294–2300. <https://doi.org/10.1157/ijaseit.8.6.6441>.
- Chen, J.K., Lee, Y.C., 2009. A new method to identify the category of the quality attribute. *Total Qual. Manag. Bus. Excell.* 20, 1139–1152. <https://doi.org/10.1080/14783360902781832>.
- Djordjevic, N., 2017. Software quality standards. *Vojnoteh. Glas.* 65, 102–124. <https://doi.org/10.5937/vojtehg65-10668>.
- Evans, I., Porter, C., Micallef, M., Harty, J., 2020. Test tools: an illusion of usability? In: 2020 IEEE International Conference on Software Testing, Verification and Validation Workshops (ICSTW). IEEE, Porto, PT, pp. 392–397. <https://doi.org/10.1109/ICSTW50294.2020.00070>.
- Hassan, H.M., Galal-Edeen, G.H., 2017. From usability to user experience. In: 2017 International Conference on Intelligent Informatics and Biomedical Sciences (ICIIBMS). IEEE, Okinawa, JP, pp. 216–222. <https://doi.org/10.1109/ICIIBMS.2017.8279761>.

- IBM-Corporation, 2014. Questionnaire template scoring formulas - ibm documentation. URL:<https://www.ibm.com/docs/en/opw/7.3.0?topic=assessments-questionnaire-template-scoring-formulas>.
- ISO/IEC, ISO 25000 STANDARDS. URL:<https://iso25000.com/index.php/en/iso-25000-standards>.
- ISO/IEC, 2011. Systems and software engineering – Systems and software Quality Requirements and Evaluation (SQUARE) – System and software quality models. URL:<https://www.iso.org/standard/35733.html>.
- ISO/IEC, 2016. Systems and software engineering – Systems and software quality requirements and evaluation (SQUARE) – Measurement of quality in use. URL: <https://www.iso.org/standard/35746.html>.
- ISO/IEC, 2019. ISO 9241-210:2019: Ergonomics of human-system interaction - Human-centred design for interactive systems. URL:<https://www.iso.org/standard/77520.html>.
- Ives, B., Olson, M.H., Baroudi, J.J., 1983. The measurement of user information satisfaction. *Commun. ACM* 26, 785–793. <https://doi.org/10.1145/358413.358430>.
- Jung, H.W., Kim, S.G., Chung, C.S., 2004. Measuring software product quality: a survey of iso/iec 9126. *IEEE Software* 21, 88–92. <https://doi.org/10.1109/MS.2004.1331309>.
- Khalifa, M., 2014. Clinical decision support: Strategies for success, in: *Procedia Comput. Sci.*, Elsevier, Halifax, CA. pp. 422–427. URL:<http://www.sciencedirect.com/science/article/pii/S187705091401028X>, doi: 10.1016/j.procs.2014.08.063.
- Kitchenham, B.A., Pfleeger, S.L., 2008. Personal opinion surveys, in: *Guid. to Adv. Empir. Softw. Eng.*. Springer, London, London, UK, pp. 63–92. URL:https://link.springer.com/chapter/10.1007/978-1-84800-044-5_3, doi: 10.1007/978-1-84800-044-5_3.
- Lew, P., Abbasi, M.Q., Rafique, I., Wang, X., Olsina, L., 2012. Using web quality models and questionnaires for web applications understanding and evaluation. In: 2012 Eighth International Conference on the Quality of Information and Communications Technology. IEEE, Lisbon, PT, pp. 20–29. <https://doi.org/10.1109/QUATIC.2012.68>.
- Lewis, J.R., 1995. IBM Computer Usability Satisfaction Questionnaires: Psychometric Evaluation and Instructions for Use. *Int. J. Human-Comput. Interact.* 7, 57–78.
- Lund, A.M., 2001. Measuring usability with the USE questionnaire. *Usability interface* 8, 3–6.
- MeasuringU, 2021. Questionnaires - measuringu. URL:<https://measuringu.com/category/questionnaires/>.
- Nakai, H., Tsuda, N., Honda, K., Washizaki, H., Fukazawa, Y., 2016. Initial framework for software quality evaluation based on iso/iec 25022 and iso/iec 25023. In: 2016 IEEE International Conference on Software Quality, Reliability and Security Companion (QRS-C). IEEE, Vienna, Au, pp. 410–411. <https://doi.org/10.1109/QRS-C.2016.66>.
- Nielsen, J., 1994. Heuristic Evaluation: How-To conduct. URL:<https://www.nngroup.com/articles/how-to-conduct-a-heuristic-evaluation/>.
- Nielsen, J., 2020. usability heuristics for user interface design. 10. URL:<https://www.nngroup.com/articles/ten-usability-heuristics/>.
- Nielsen, J., Landauer, T.K., 1993. A mathematical model of the finding of usability problems, in: *Proceedings of the INTERACT '93 and CHI '93 Conference on Human Factors in Computing Systems*, Association for Computing Machinery, New York, NY, USA. p. 206–213. doi: 10.1145/169059.169166.
- P. Miguel, J., Mauricio, D., Rodríguez, G., 2014. A Review of Software Quality Models for the Evaluation of Software Products. *Int. J. Softw. Eng. Appl.* 5, 31–53. doi: 10.5121/ijsea.2014.5603, arXiv:1412.2977.
- Pavlič, L., Okorn, M., Heričko, M., 2018. The use of the software metrics in practice. *CEUR Workshop Proc.* 2217, 27–30.
- Pincioli, F., 2016. Improving Software Applications Quality by Considering the Contribution Relationship among Quality Attributes. *Procedia Comput. Sci.* 83, 970–975. <https://doi.org/10.1016/j.procs.2016.04.194>. URL:<https://doi.org/10.1016/j.procs.2016.04.194>.
- Sacha, K., 2006. Evaluation of expected software quality: A customer's viewpoint. *Lect. Notes Comput. Sci. (including Subser. Lect. Notes Artif. Intell. Lect. Notes Bioinformatics)* 3922 LNCS, 170–183. doi: 10.1007/11693017_14.
- van Solingen, R., Berghout, E., 1998. *The Goal/Question/Metric Method: a practical guide for quality improvement of software development*, volume 7. McGraw-Hill Education. URL:https://www.academia.edu/26126148/The_Goal_Question_Metric_method_a_practical_guide_for_quality_improvement_of_software_development.
- Sommerville, I., 2016. *Software engineering*. Addison Wesley, New York, USA.
- Souza-Pereira, Leonice, Pombo, Nuno, Ouhbi, Sofia, 2021a. Quality-in-use Characteristics for Clinical Decision Support System Assessment. *Comput. Methods Programs Biomed.* 207. <https://doi.org/10.1016/j.cmpb.2021.106169>.
- Souza-Pereira, Leonice, Pombo, Nuno, Ouhbi, Sofia, 2021b. A process model for quality in use evaluation of clinical decision support systems. *J. Biomed. Inf.* 123. <https://doi.org/10.1016/j.jbi.2021.103917>.
- Souza-Pereira, L., Pombo, N., Ouhbi, S., Felizardo, V., Garcia, N., 2020. Clinical decision support systems for chronic diseases: A Systematic literature review. *Comput. Methods Programs Biomed.* 195. <https://doi.org/10.1016/j.cmpb.2020.105565> 105565.
- Sulla-Torres, J., Gutierrez-Quintanilla, A., Pinto-Rodriguez, H., Gómez-Campos, R., Cossio-Bolaños, M., 2020. Quality in use of an android-based mobile application for calculation of bone mineral density with the standard ISO/IEC 25022. *Int. J. Adv. Comput. Sci. Appl.* 11, 158–163. <https://doi.org/10.14569/IJACSA.2020.0110821>.
- Tsuda, N., Washizaki, H., Honda, K., Nakai, H., Fukazawa, Y., Azuma, M., Komiya, T., Nakano, T., Suzuki, H., Morita, S., Kojima, K., Hando, A., 2019. Wsqf: Comprehensive software quality evaluation framework and benchmark based on square. In: 2019 IEEE/ACM 41st International Conference on Software Engineering: Software Engineering in Practice (ICSE-SEIP). IEEE, Montreal, CA, pp. 312–321. <https://doi.org/10.1109/ICSE-SEIP.2019.00045>.
- Zikos, D., 2017. A Framework to Design Successful Clinical Decision Support Systems, in: *Proc. 10th Int. Conf. Pervasive Technol. Relat. to Assist. Environ.*, Association for Computing Machinery, New York, NY, USA. pp. 185–188. URL: <https://doi.org/10.1145/3056540.3064960>, doi: 10.1145/3056540.3064960.